

Nicholas Swisher

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SUMMARY

New Grad Software Engineer with experience building and deploying API-driven applications, applying strong CS fundamentals, and actively expanding into modern JavaScript based full-stack development.

EDUCATION

University of Colorado Boulder <i>B.S. Applied Computer Science — GPA: 3.74 — Cum Laude</i>	Boulder, CO Dec 2025
University of North Texas <i>B.S. Kinesiology — GPA: 3.6 — Cum Laude</i>	Denton, TX May 2020

PROJECTS

Content Recommendation System

Python, FastAPI, SQLite

- Built a content-based recommendation engine using TF-IDF and cosine similarity, exposing results through FastAPI REST endpoints.
- Designed a reusable feature pipeline backed by SQLite and deployed the FastAPI service on Render for integration into user-facing applications.

Event Management Platform

C++

- Developed a modular backend system modeling users, events, and transaction workflows using object-oriented design.
- Applied Factory and Singleton patterns to support extensibility and centralized application coordination.

Data Structures and Benchmarking

C++

- Implemented templated HashMap, BST, and Linear List structures with full Catch2 unit test coverage.
- Benchmarked runtime performance across datasets to evaluate algorithmic tradeoffs and guide optimization decisions.
- Integrated continuous testing with GitHub Actions CI to automatically validate correctness across changes.

TECHNICAL SKILLS

Languages: C++, Python, SQL

Frameworks & Databases: FastAPI, scikit-learn, pandas, SQLite, MongoDB

Concepts: OOP, Data Structures & Algorithms, Design Patterns, REST APIs, Unit Testing (Catch2), Test-Driven Development

Tools: Git, GitHub Actions, VS Code

EMPLOYMENT

Safeway

Front End Manager

2024 – 2025

- Led front-end operations, resolving customer issues and coordinating team workflows.

Spectrum

Account Manager

2023 – 2024

- Managed client communication and issue resolution while supporting account operations.

Rocky Mountain Cycle Plaza

Technician

2021 – 2023

- Diagnosed and repaired complex mechanical systems while communicating findings to customers.